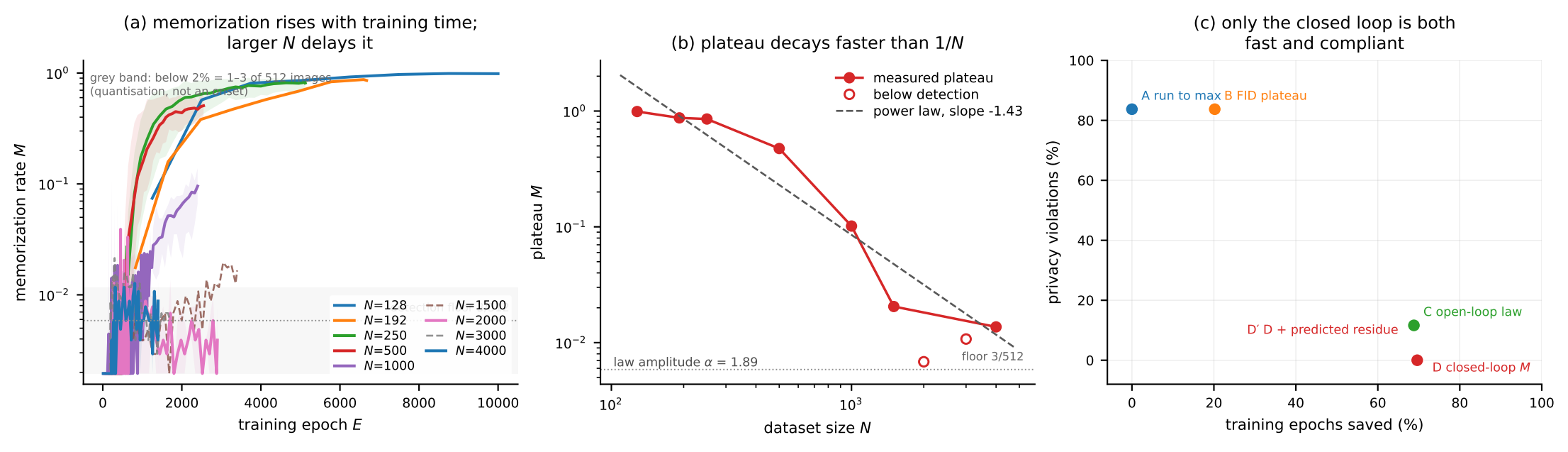
(a) memorization rises with training time; larger N delays it
grey band: below 2% = 1–3 of 512 images (quantisation, not an offset)
memorization rate M
training epoch E
N=128
N=192
N=250
N=500
N=1000
N=1500
N=2000
N=3000
N=4000
(b) plateau decays faster than 1/N
measured plateau
below detection
power law, slope -1.43
plateau M
dataset size N
law amplitude α = 1.89
floor 3/512
(c) only the closed loop is both fast and compliant
privacy violations (%)
training epochs saved (%)
A run to max
B FID plateau
C open-loop law
D′ D + predicted residue
D closed-loop M